Phase 1 Final Assessment

Local AI Support Assistant

Practical readiness assessment for Phase 2: LLM Application Development

Assessment Session Log

Start date	**October 25, 2026**
Start time	12:55 PM PHT (UTC+8)
End time	Record when assessment is finished
Target duration	3–5 hours

Purpose: Build a small working application that accepts a technical question, sends it to a local Ollama model, and returns a response through an HTTP API. This assesses Python fundamentals, API development, validation, error handling, testing, and project hygiene.

1. Mission

Build a Local AI Support Assistant. Example use case: a developer reports that an API returns HTTP 500. Your application should accept the question and return troubleshooting suggestions.

Scope boundary: Do not build RAG, vector database functionality, LangGraph workflows, or advanced agents. The goal is to demonstrate Phase 1 foundations.

2. Requirements and Scoring Rubric

Category	Points	What to demonstrate
Project setup	10	Use a virtual environment, organize files into modules, document installation/run steps, and keep secrets/local configuration out of Git.
Python fundamentals	15	Use reusable functions, appropriate data structures, type hints, and separate API, model-client, and configuration responsibilities.
Request validation	15	Implement POST /chat with Pydantic request/response models. Reject empty prompts and prompts over 2,000 characters.
Ollama integration	20	Send the validated prompt to a local Ollama model and return its response. Do not hard-code the model name throughout the application.
Error handling and logging	15	Handle model-service outages and model errors without crashing. Return suitable HTTP errors and log useful diagnostics without unnecessarily logging sensitive prompts.
Automated testing	15	Use pytest to test valid requests,

		invalid prompts, and model-service failures. Mock the model call so tests do not require Ollama to run.
Documentation and Git	10	Write a README with setup, configuration, run instructions, endpoint examples, and test commands. Commit the work to a dedicated Git branch.

Total: 100 points. The final score will be based on the actual implementation, not simply checking boxes.

3. Suggested Project Structure

```
phase1-final-assessment/
├── app/
│   ├── __init__.py
│   ├── main.py
│   ├── schemas.py
│   ├── config.py
│   └── llm_client.py
├── tests/
│   ├── __init__.py
│   └── test_chat.py
├── .env.example
├── .gitignore
├── README.md
└── requirements.txt
```

4. API Contract

Endpoint: POST /chat

Request example:

```
{
  "prompt": "My API returns HTTP 500. How do I troubleshoot it?"
}
```

Expected response format (model wording may differ):

```
{
  "response": "Check the application logs, reproduce the error, and inspect the server-side exception..."
}
```

5. Required Behaviors

Scenario	Expected behavior
Valid prompt	Return HTTP 200 with a response.
Empty prompt	Return HTTP 422.
Prompt over 2,000 characters	Return HTTP 422.
Ollama unavailable	Return an appropriate 5xx response.
Model returns an error	Return a controlled error response.
Automated tests	Pass without requiring Ollama to be running.

6. Rules

- Official documentation and syntax searches are allowed.
- Try implementing the solution yourself before asking AI to explain an error.
- Do not copy an entire ready-made project.
- Do not start Phase 2 topics such as RAG, vector databases, LangGraph, or agent frameworks.
- The code must run locally, and you must be able to explain how it works.
- Pydantic is required; Pydantic AI is not required.

7. Readiness Decision

80–100 points — Ready for Phase 2	**Fundamentals are strong enough to begin LLM application development.**
60–79 points — Almost ready	Address weak areas with targeted improvements before moving on.
Below 60 points — Strengthen Phase 1	Practice the specific skills that need improvement.

Regardless of score, a working endpoint, valid request validation, and basic automated tests are required before recommending progression.

8. Final Submission Checklist

☐ Project file structure

☐ Contents of main.py, schemas.py, config.py, and llm_client.py

☐ Test file and output of pytest

☐ One successful API response and one error response

☐ README with setup and usage instructions

9. Mentor Review

Final score: _____ / 100

Decision: ☐ Pass ☐ Conditional pass ☐ Retry

Strengths: _____

Areas to improve: _____

Next steps: _____

10. Session Completion

End date: _____

End time (PHT, UTC+8): _____

Total duration: _____

When ready for review, share the project files or upload a ZIP after removing real secrets. The mentor review will score each category, explain issues, and recommend whether to proceed to Phase 2.